

- Other characteristics of a RISC architecture
 - 1) A relatively large number of registers in the processor unit
 - 2) Use of overlapped register windows to speed-up procedure call and return
 - 3) Efficient instruction pipeline
 - 4) Compiler support for efficient translation of high-level language programs into machine language programs

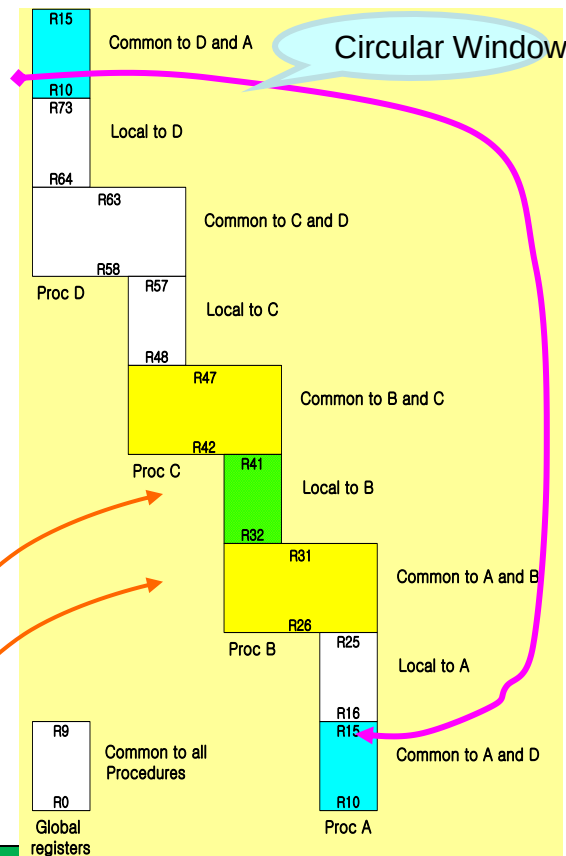
– Overlapped Register Windows

- Time consuming operations during procedure call
 - Saving and restoring registers
 - Passing of parameters and results
- Overlapped Register Windows
 - Provide the passing of parameters and avoid the need for saving and restoring register values **by hardware**

– Concept of overlapped register windows : Fig. 8-9

- Total 74 registers : R0 - R73

- R0 - R9 : Global registers
 - R10 – R73 : 4 windows
 - » Window A
 - » Window B
 - » Window C
 - » Window D
- 10 Local registers**
+
2 sets of 6 registers
(common to adjacent windows)



- **Example**) Procedure A calls procedure B

- R26 - R31
 - » Store parameters for procedure B
 - » Store results of procedure B
- R16 - R25 : Local to procedure A
- R32 - R41 : Local to procedure B

G : Global registers = 10

L : Local registers = 10

C : Common registers = 6

W : Number of windows = 4

- Window Size = $L + 2C + G = 10 + (2 \times 6) + 10 = 32$ registers
- Register File (total register) = $(L + C) \times W + G = (10 + 6) \times 4 + 10 = 74$ registers

